Transfer: circuits and induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply circuit relationships and explain an induction observation.

SUCCESS CRITERIA - I can...

1. Use current and voltage rules in a new circuit.
2. Account for electrical power and energy.
3. Explain induction through changing flux.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “resistance” and “power” to explain your model or result. Sentence starters are optional.

resistance: voltage divided by current for a stated operating condition

power: rate of energy transfer

induction: producing an emf through a changing magnetic flux

01 Warm-up: bring back an earlier idea

A device receives 50 J mechanical energy at its shaft and produces electrical energy at its terminals.
Identify it and name one possible loss.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $V = IR$; $P = IV$; $E = P\Delta t$; $|\text{emf}_{avg}| = N|\Delta\Phi|/\Delta t$.
Use stated circuit or coil conditions. Ideal battery and wires unless stated otherwise.

Worked reasoning: A 12 V supply drives a 6 ohm resistor: current 2 A and power 24 W. A complete model checks both the charge-flow rate and the energy-transfer rate.

Your turn: Draw the circuit and an energy-flow arrow. **Label** what an ammeter and voltmeter would measure and where each connects.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch two resistors in parallel across a battery and label the equal-voltage branches.

Sentence starter: The … in my model represents … . Its label shows … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A 6 V battery powers 3 Ω and 6 Ω resistors in parallel.
Label both branch voltages.
Find each current, total source current and total power $P = VI_{total}$.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A magnet is held motionless near a fixed coil, then moved into it.
Explain the change in the meter response without saying the magnet supplies unlimited energy.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 Ω resistor carries 6 A.
Find the potential difference across it.
Draw the stated circuit and label quantities relevant to the question.

Sentence starter: Both … . They differ in … because … .

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

A motor uses 6 W for 5 s.
Find energy and name one possible non-mechanical output.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Replace the resistor with two 12 ohm resistors in parallel.
Compare total current and power, then draw both branch currents.

Sentence starter: Both … . They differ in … because … .



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3